

SANKET VIRANI

Birthdate: 04.04.1997 | Nationality: Indian

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🏠: Wolfsburg, Germany



Familiar with FEA/FEM-based electronic simulations using ANSYS and handling CAD/ECAD/ODB++ data for system-level analysis. Possesses a strong understanding of technical project coordination, focusing on accelerated lifetime testing and reliability evaluation of automotive electronics based on mission profiles. Experienced in electronics reliability, degradation mechanisms, failure analysis, and lifetime assessment of components.

SKILLS

Technical Project Coordination
ANSYS Sherlock/Mechanical, LT-Spice/ AutoCAD
FEA /FEM / CAE / CAD / ODB++ Data Formats
Simulation-Based Validation and Reliability Modelling
Electronics Reliability, Robustness & Failure Analysis
Microsoft 365

Electronics Tools (Multimeter, Soldering tools, Oscilloscope)
PCB Assembly, Testing, and Troubleshooting
Semiconductor & Power Electronics Testing
Business English (C1) & German(A2+)

WORK EXPERIENCE

INTERNSHIP

Virtual Lifetime Simulation of Automotive Electronics Components

09/2024 – 03/2025 (Volkswagen AG – Wolfsburg, Germany)

- Working on electronics FEA & FEM based reliability simulation methodologies with ANSYS tools
- Conducting automotive E/E component (ECU/Infotainment display, PCBA) failure analysis in CAD/CAE tools
- Experienced and knowledge of ODB++, CAD/CAE data format and modified according to the requirements
- Simulated reliability and lifetime of automotive electronics using physics-based models and stress analysis
- Developing thermal, mechanical and electrical simulation models of electronics based on Mission Profile
- Conducting literature reviews on virtual simulation and providing technical guidance to the team members

Tools & Standards: - MS Office, ANSYS Tools, LV124 (VW80000), ODB++/CAD/CAE/HMI

MASTER THESIS

Accelerated Lifetime Testing for Evaluating Robustness in Automotive Electronics

01/2024 – 09/2024 (Volkswagen AG – Wolfsburg, Germany)

- Collaborating with suppliers (Tier-1/2...) & researched the impact of emerging trends on automotive extended lifetime/reliability testing of electronic components
- Worked with OEMs and Tier-1 suppliers to ensure compliance with automotive standards (AEC-Q100, JEDEC, VW80000)
- Optimization of testing procedures and derive effective solutions based on future-oriented project
- Developed/validated Thermal Performance Measurement (TPM)-based electronics Qualification Flowchart
- Supporting in the development and validation process of Infotainment system projects (HMI)
- Utilized climate chambers for accelerated stress tests (HTOL, TC, HTSL) on infotainment display
- Researched on Time to Failures (TTF), degradation and lifetime of automotive electronics

Tools & standards: - MS Office, VW80000, AEC-Q, JEDEC, ZVEI, IEEE, MP,HMI, Climate Chamber

INTERNSHIP

Reliability and Accelerated Lifetime of Electronics

06/2023 – 12/2023 (Volkswagen AG – Wolfsburg, Germany)

- Performed Service Life Tests (L-02/HTOL) and evaluated critical test groups for extended lifetime impacts
- Reviewed relevant documents on electronics robustness and lifespan, referring to JEDEC, AEC-Q, ZVEI standards and presented and discussed with team members
- Created questionnaires for Semiconductor supplier discussions on extended lifetime based on Mission Profile (MP) of electronics components
- Collaborated with suppliers and manufacturers to grasp electronic devices failure behavior and degradation trends
- Discussed automotive test standards with Tier-1/2, to improve reliability and lifetime testing standards

Tools & Standards: - MS Office, VW80000, AEC-Q, JEDEC, ZVEI, IEEE, HTOL, TC, HTSL, Climate Chamber

FULL TIME Electronics Engineer

03/2019 – 12/2021 (Meghal Laser Process PVT LTD – Surat, India)

- Proficient in maintaining and troubleshooting electronic equipment and components to ensure optimal performance
- Experienced in using electronic tools and equipment for hands-on problem-solving in electronics field
- Managed service, maintenance, and production departments, optimizing design and ordering processes
- Skilled in PCBA implementing, and testing circuits for electronic devices and quality control

Tools: - MS Office, Multimeter, Soldering Tools, Oscilloscopes, LT- SPICE, RCA, Other Electronics Tools

EDUCATION & PROJECTS

03/2022 – 09/2025 M.E. in Electrical and Microsystems Engineering (OTH Regensburg, Germany)

- **Key Subjects:** Advanced Semiconductor Technologies and Devices, Electronics Product Engineering, LED Technology, Project Management, Advance Mathematics, Engineering Chemistry

Project Detecting Periodicity of infinite unknown data with FFT

- Determined the periodicity of the countless unknow data with specific tolerances with the help of FFT (Fast Fourier Transfer) using mathematical equation in C++ and MATLAB

Tools: - C++, MATLAB, GNU-Plot, MS Excel, Notepad

06/2014 – 06/2018 B.E. in Electronics and Communication Engineering (GTU– Ahmedabad, India)

- **Key Subjects:** Analog and Digital Electronics, Power Electronics, Electronic Devices and Circuits, Circuit Network, Analog, Digital and Wireless Communications, Advance Mathematics

Project Automated Toll Plaza using RFID

- Designed and implemented an Automated Toll Plaza system using RFID based technology to improved traffic flow and reduced congestion at the toll plaza

Tools: - C language, Microcontroller AT89s52, RFID sensors or Tag, Step Motor

HOBBIES, PROFESSIONAL SUMMARY & OTHERS

Hobbies Reading Technical Documents, Yoga/Meditation, Team Sports

Professional summary: Self-motivated with the ability to prioritize and manage time effectively
Proven ability to rely on limited experience and judgment to plan and achieve objectives
Exceptional planning, analytical, and decision-making skills

Other PKW – Class B (Driving License)

Sanket Virani
Signature